ObjDream: Guiding Vision-Language-Action Models with Object-Level Understanding and Future State Prediction

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https://huailiangma.github.io/objdream/

Abstract: Vision-language-action (VLA) models have emerged as a powerful paradigm for robotic manipulation, enabling robots to generate actions from visual observations and language instructions. However, most existing VLA models still operate on image-level visual representations inherited from pretrained vision-language backbones, while lacking fine-grained object understanding for manipulation. As a result, their robustness can degrade under out-of-distribution object layouts, clutter, and background variations. To address these limitations, we introduce ObjDream, an object-centric VLA framework centered on object-level understanding and future object state prediction. ObjDream uses a spatio-temporal object encoder to organize visual observations into temporal object slots, where persistent object identity and dynamic object state are represented through separate channels. These structured object representations are inserted into the token sequence of the vision-language model (VLM), where identity-guided routing modulates VLM attention to the corresponding object tokens. In addition, an action-conditioned object dynamics module predicts compact future object states relevant to manipulation during training, explicitly supervising object state evolution under robot actions without requiring future prediction at inference. Experiments on LIBERO, LIBERO-Plus, and RoboTwin 2.0 show that ObjDream improves generalization under diverse distribution shifts and scene perturbations. Real-world robot experiments further demonstrate that ObjDream transfers to physical manipulation settings with novel object arrangements and background changes.

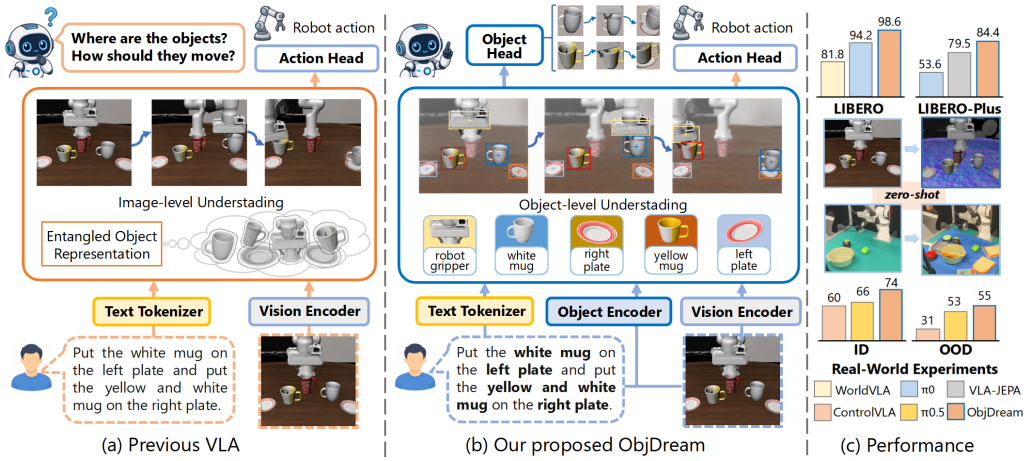


Figure 1: **Overview of ObjDream.** We introduce ObjDream, an object-centric VLA framework for robust manipulation. Unlike prior VLA models (a), which primarily rely on image-level understanding and entangled object representations, ObjDream (b) introduces explicit object-level representations and future object state prediction. This object-centric formulation improves manipulation robustness, leading to stronger performance than baseline methods (c).

1 Introduction

Robust robot policies must reliably manipulate objects in out-of-distribution environments, where their appearance, position, and surrounding context may differ from those in the training data. VLA models have made rapid progress in generalist robot control by scaling pretrained vision-language backbones, robot data, and action decoders (Brohan et al., 2023; Kim et al., 2024; Black et al., 2024; NVIDIA et al., 2025; Wang et al., 2026b). To improve generalization, recent work has sought to preserve knowledge acquired during pretraining (Guo et al., 2026; Hancock et al., 2026), use high-level plans to guide low-level control (Li et al., 2025c), or anticipate scene changes through future visual or latent-state prediction (Zhang et al., 2025; Sun et al., 2026). Although these approaches improve generalization to some extent, most VLA models still lack fine-grained object-level understanding. Their visual representations, inherited from VLMs, remain largely image-level, leaving the identity and spatial state of task-relevant objects entangled with appearance and surrounding context. This entanglement can weaken out-of-distribution robustness under shifts in object appearance, spatial layout, and background context.

Object-centric approaches offer a natural way to strengthen object-level understanding by explicitly representing individual objects and their properties. Prior work incorporates object-aware visual representations to help policies focus on task-relevant regions (Zhu et al., 2023; Li et al., 2025a; Liang et al., 2026). Other methods extend object representations to relational reasoning (Hanyu et al., 2025), object-level memory (Chung et al., 2026), and long-horizon manipulation (Yang et al., 2026). Although these approaches demonstrate the value of object-level representations, they often encode persistent object identity and time-varying object state in the same slot or token. As appearance and geometry change, this coupling can obscure persistent identity cues by mixing them with dynamic object state. Recent dual-state slot models show that separating identity from dynamic appearance can improve temporal consistency in video perception by keeping identity information distinct from time-varying visual content (Tran et al., 2026). Building on this insight, we investigate whether persistent identity references can be represented alongside dynamic object states and further used to guide VLM attention for language-conditioned manipulation.

Beyond modeling the current visual observation, predictive models support manipulation by anticipating future observations or object states (Ferraro et al., 2025; Chen et al., 2025a; Villar-Corrales and Behnke, 2025). The choice of prediction target is important: reconstructing complete images requires modeling texture and background details that may be irrelevant to control (Zhang et al., 2025), while goal-image methods such as Goal-VLA introduce an additional synthesis stage before estimating the target object pose (Chen et al., 2025a). In contrast, object-centric world models show that scene evolution can be modeled using compact object representations (Jeong et al., 2025). Motivated by this trade-off, we use action-conditioned prediction of compact future object states, including geometry and visibility, as auxiliary supervision for VLA training without requiring future object state prediction at inference.

To this end, as illustrated in Fig. 1, we introduce ObjDream, an object-centric vision-language-action framework for improving manipulation generalization. The architecture of ObjDream builds on a pretrained VLM backbone and integrates a spatio-temporal object encoder, an object-level forward dynamics module, and a flow-matching action expert. Specifically, the spatio-temporal object encoder transforms visual observations into structured object slots, where persistent object identity and dynamic object state are represented through separate channels. These factorized object representations are mapped into the VLM hidden space and inserted into the token sequence, where identity-guided routing modulates attention to the corresponding object tokens. The forward dynamics module predicts future object states during training, providing auxiliary supervision for future object state evolution under robot actions without requiring future prediction at inference. The object encoder is pretrained with automatically generated offline object supervision and frozen during VLA training. We optimize ObjDream with a joint objective combining action imitation with current and future object state supervision. Extensive evaluations on LIBERO, LIBERO-Plus, RoboTwin 2.0, and real-world robot experiments demonstrate competitive manipulation performance across diverse scene perturbations.

The main contributions of this paper are listed below:

- We present ObjDream, an object-centric VLA framework centered on object-level understanding and future object state prediction to improve generalization under perturbations. ObjDream

introduces a spatio-temporal object encoder that explicitly represents persistent object identity and dynamic object state through separate channels. The resulting representations are integrated into the VLM hidden space with a novel identity-guided routing mechanism for robust language-conditioned manipulation.

- We introduce an action-conditioned object-level forward dynamics module for modeling object evolution during manipulation. It predicts compact future object states during training to provide direct auxiliary object-level supervision without requiring future object state prediction at inference. We further model inter-object relations from object features and pairwise geometry to provide structured context for action generation.
- Extensive evaluations on LIBERO, LIBERO-Plus, RoboTwin 2.0, and real-world robot experiments demonstrate competitive manipulation performance across diverse scene and environmental perturbations. Ablations further assess the effects of the key architectural components.

2 Related Work

Vision-language-action models. VLA models combine pretrained language or vision-language backbones with action decoders to learn general-purpose manipulation from large-scale robot demonstrations (Brohan et al., 2022, 2023; O'Neill et al., 2024; Team et al., 2024; Kim et al., 2024; Black et al., 2024; Physical Intelligence et al., 2025). Recent work improves VLA performance through stronger action decoders and fine-tuning strategies (Chi et al., 2025; Zhao et al., 2023; Kim et al., 2025; Hou et al., 2025; NVIDIA et al., 2025; Qu et al., 2025), larger-scale cross-embodiment training (Cai et al., 2026; Team et al., 2026; Wu et al., 2026a,b; Wang et al., 2026b), and better preservation or exploitation of pretrained visual-semantic and spatial priors (Guo et al., 2026; Li et al., 2026a; Wang et al., 2026c). Predictive objectives further use future images, video tokens, subgoal images, or latent states to provide temporal supervision for manipulation (Zhao et al., 2025; Zhang et al., 2025; Cen et al., 2025b,a; Li et al., 2026b). Despite these advances, existing VLA models still largely operate on dense image-level visual representations, where object-level structure is not explicitly organized for manipulation generalization. ObjDream addresses this limitation by enhancing VLA policies with object-level understanding and future object state supervision.

World models for robot control. World models learn action-conditioned environment dynamics and support robot control by planning or learning policies from predicted futures in observation or latent spaces (Hafner et al., 2019; Wu et al., 2023). Recent work extends this paradigm through language-conditioned video generation (Du et al., 2023; Zhou et al., 2024b), pretrained visual latent dynamics (Zhou et al., 2024a; Assran et al., 2025), and world-action or unified video-action models that couple future prediction with executable actions (Ye et al., 2026; Li et al., 2026c). These approaches provide dynamics supervision by modeling anticipated scene evolution. However, prediction in pixel or video space requires modeling high-dimensional visual details and can introduce training or inference overhead. Recent approaches therefore explore more efficient formulations, including avoiding explicit future imagination at inference or exposing compact representations of scene changes (Yuan et al., 2026; Bai et al., 2026). ObjDream follows this direction by using compact future object state prediction as supervision during training, while directly generating actions at inference without explicit future object state decoding.

Object-centric representations for manipulation. Object-centric representation learning decomposes visual scenes into entity-level latent variables and has been widely studied in scene decomposition, temporal tracking, and structured dynamics (Locatello et al., 2020; Kipf et al., 2021; Wu et al., 2022). In robotic manipulation, object-level representations have been used to improve sample efficiency and enhance visuomotor attention and robust policy conditioning through object proposals, grounded geometry, object-centric prompting, and physically grounded object concepts (Zhu et al., 2023; Vo et al., 2025; Li et al., 2025b; Wei et al., 2026), enable efficient VLA learning through compact object-centric tokenization (Bendikas et al., 2025), and support structured control through object-centric dynamics and relational representations (Ferraro et al., 2025; Mosbach et al., 2024; Hanyu et al., 2025; Yang et al., 2026). Complementary approaches use structured affordance representations to provide action-relevant object grounding, interaction localization, and geometric guidance for VLA policies (Yu et al., 2026b). Recent predictive approaches further incorporate object-aware future representations, masks, or addressable slot states into world-action learning (Wang et al., 2026a; Yu et al., 2026a; Liu et al., 2026). Recent dual-state slot models separate per-frame appear-

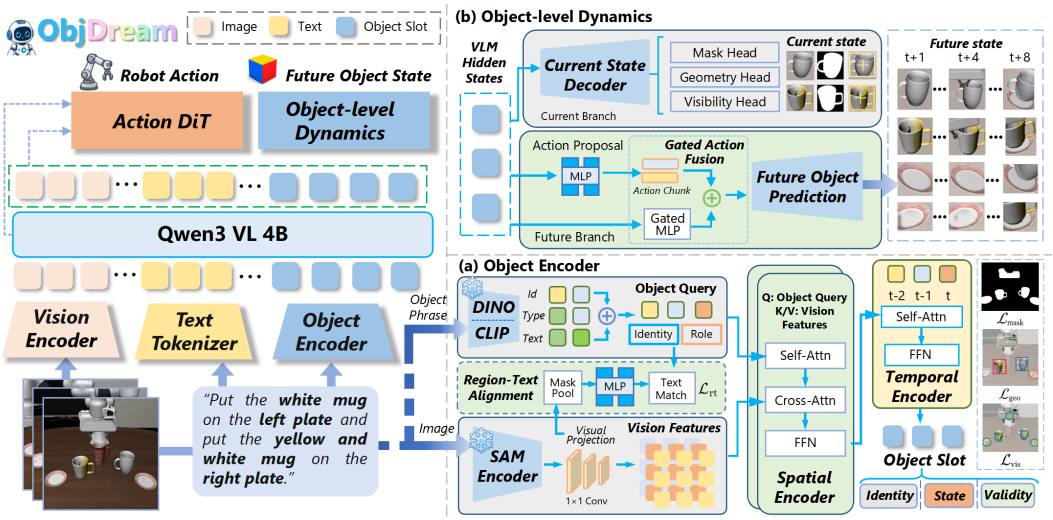


Figure 2: **Overview of the ObjDream architecture.** The model integrates three main components: (1) a spatio-temporal object encoder that constructs factorized object slots by separately representing persistent object identity and time-varying object state; (2) an object-level dynamics module that decodes current object states and predicts action-conditioned future object states; and (3) a flow-matching action expert that predicts robot action chunks. ObjDream enhances object-level understanding through factorized object representations and future object state modeling, enabling more generalizable language-conditioned manipulation under object and scene variations.

ance from persistent object identity, highlighting the value of disentangling identity from changing visual content in object-centric representations (Tran et al., 2026). From this perspective, ObjDream combines persistent identity references with dynamic object states and uses identity-guided routing to provide object-specific context for language-conditioned manipulation.

3 Method

Following the VLA policy formulation in Sec. 3.1, Fig. 2 illustrates the architecture of ObjDream. ObjDream combines a spatio-temporal object encoder, an object dynamics module, and a flow-matching action expert. The encoder produces structured object slots that are mapped to VLM object tokens, while identity-guided routing uses cached identity references to modulate attention to these tokens (Sec. 3.2). The object dynamics module provides future object state supervision during training (Sec. 3.3), and the action expert generates action chunks.

3.1 VLA Policy Formulation

We formulate ObjDream as an object-centric VLA policy that predicts continuous action chunks from language instructions, visual observations, proprioceptive states, and object-level embeddings. At control step t, the policy receives a language instruction x, a current multi-view observation I_t, and proprioceptive state s_t. The VLM consumes all policy views in I_t, while the object encoder operates on a designated external or third-person view with causal history $\mathbf{I}_t = (I^{\text{enc}}_{t-T+1}, \ldots, I^{\text{enc}}_t)$. At episode initialization, a phrase parser extracts a task-relevant phrase set P_0 from the instruction x and initial observation I_0. A spatio-temporal object encoder E_ω then produces structured object slots $O_t = E_\omega(\mathbf{I}_t, P_0)$. The VLA backbone f_ψ encodes these inputs into a multimodal context C_t, and the action expert parameterizes the conditional distribution over an action chunk $A_t = [a_t, \ldots, a_{t+H_a-1}]$:

$$C_t = f_\psi(x, I_t, s_t, O_t), \qquad p_\theta(A_t \mid x, I_t, s_t, O_t) = p_\theta(A_t \mid C_t). \tag{1}$$

Here, H_a denotes the action horizon. In our implementation, p_θ is instantiated by a flow-matching action expert (Lipman et al., 2022; Black et al., 2024). Given a demonstrated action chunk $A^\star_t$, Gaussian noise $Z_t \sim \mathcal{N}(0, I)$, and flow time $\tau \sim \mathcal{U}[0, 1]$, the action loss is

$$A^\tau_t = (1-\tau)Z_t + \tau A^\star_t, \qquad \mathcal{L}_{\text{act}} = \mathbb{E}\left[\left\|v_\theta(A^\tau_t, \tau; C_t) - (A^\star_t - Z_t)\right\|^2_2\right]. \tag{2}$$

At inference, the learned velocity field is integrated to generate an action chunk, which is executed in a receding-horizon manner.

3.2 Object-Centric Tokenization and VLM Routing

Spatio-temporal object encoder. ObjDream uses a spatio-temporal object encoder E_ω to construct factorized object slots from the causal image window $\mathbf{I}_t$ and the cached phrase set P_0. At episode initialization, Qwen3-VL-8B (Bai et al., 2025) processes the instruction x and initial observation I_0 once to extract task-relevant object phrases, which are cached for all subsequent control steps. Query ordering is fixed across frames, with reserved robot queries and padding to N_s slots. For each phrase p_i, a frozen CLIP text encoder (Radford et al., 2021) produces f_i, and a learnable projection gives $\iota_i = \mathrm{Normalize}(P_{\mathrm{sem}}(f_i))$. The object query is initialized as

$$q_i^{(0)} = P_{\mathrm{id}}(\iota_i) + P_{\mathrm{text}}(f_i) + E_{\mathrm{type}}(\chi_i), \tag{3}$$

where $P_{\mathrm{id}}, P_{\mathrm{text}}$ are learnable projections and E_{type} embeds the query type χ_i. A frozen DINOv2 encoder (Oquab et al., 2023) extracts a visual reference d_i from the initial object region predicted by the encoder, which is cached throughout the episode. Robot queries use learned identities and type priors without the direct text term.

A spatial query module applies self-attention and cross-attention to learnably projected features from a frozen SAM2 image backbone (Ravi et al., 2025), followed by causal temporal encoding. The encoder outputs

$$O_t = E_\omega(\mathbf{I}_t, P_0) = \left\{u_{i,t}^{\mathrm{obj}}\right\}_{i=1}^{N_s}, \qquad u_{i,t}^{\mathrm{obj}} = (e_i^{\mathrm{id}}, e_{i,t}^{\mathrm{state}}, \nu_{i,t}). \tag{4}$$

The persistent identity $e_i^{\mathrm{id}} = [\iota_i; d_i; r_i]$ combines semantic and visual references with a role embedding r_i, while $e_{i,t}^{\mathrm{state}} = [c_{i,t}^{\mathrm{app}}; g_{i,t}^{\mathrm{geo}}; \tau_t]$ encodes dynamic appearance, geometry, and temporal information. Slot validity $\nu_{i,t}$ masks inactive or padded slots and is distinct from predicted visibility. During pretraining, regional features obtained by mask pooling from the projected SAM2 features, extracted before query cross-attention, are aligned with frozen CLIP text features. The objective is described in Sec. 3.4.

Factorized object injection. We map identity and dynamic state components separately into the VLM hidden space:

$$h_{i,t}^{\mathrm{obj}} = \left[P_{\mathrm{id}}^{\mathrm{inj}}(e_i^{\mathrm{id}}); P_{\mathrm{obj}}\left([P_{\mathrm{state}}(e_{i,t}^{\mathrm{state}}); \phi_\nu(\nu_{i,t})]\right)\right]. \tag{5}$$

Here, $P_{\mathrm{id}}^{\mathrm{inj}}$, P_{state}, and P_{obj} are lightweight projections, and ϕ_ν embeds slot validity. The factorization is imposed at the encoder output and initial object-token construction; subsequent VLM contextualization is unrestricted, while the cached identity reference e_i^{id} remains separate for routing. Attention keys and values are computed from the full contextualized token.

Relation and prospective object tokens. Beyond individual object tokens, ObjDream introduces a relation token and $N_p = 2$ prospective object tokens:

$$h_t^{\mathrm{rel}} = \Phi_{\mathrm{rel}}\left([\mathrm{Pool}_i(h_{i,t}^{\mathrm{obj}}); \mathrm{Pool}_{i,j}(\Delta g_{ij,t})]\right), \qquad h_{k,t}^{\mathrm{pro}} = \Phi_{\mathrm{pro}}\left([h_{i_k,t}^{\mathrm{obj}}; h_t^{\mathrm{rel}}]\right). \tag{6}$$

Here, Φ_{rel} aggregates object features and pairwise geometry into a compact global relational context, while Φ_{pro} combines this context with individual object features. Pool denotes aggregation over valid slots or valid slot pairs, and $\Delta g_{ij,t}$ encodes the relative geometry between slots i and j. The indices i_k correspond to the top-N_p valid slots ranked by a learned slot importance gate. These prospective tokens are optimized through the action objective to provide object-specific context for action generation. Together, the tokens form the object block

$$X_t^{\mathrm{obj}} = [h_{1,t}^{\mathrm{obj}}, \ldots, h_{N_s,t}^{\mathrm{obj}}, h_t^{\mathrm{rel}}, h_{1,t}^{\mathrm{pro}}, \ldots, h_{N_p,t}^{\mathrm{pro}}]. \tag{7}$$

The block is inserted between the multimodal prefix and the action-generation prompt. The attention mask retains causal attention and additionally permits bidirectional attention within the object block, excluding invalid and padded slots as keys.

Identity-guided object routing. For the base object token associated with slot i at VLM position j_i, query position p, layer ℓ, and attention head m, let $\bar{h}_p^\ell$ denote the normalized query hidden state. Using the cached identity reference e_i^{id}, we compute an identity matching score b_{pi}^ℓ and a head-specific gate $\eta_p^{\ell,m}$:

$$b_{pi}^\ell = \cos\big(U^\ell \bar{h}_p^\ell, A^\ell e_i^{\mathrm{id}}\big), \qquad \eta_p^{\ell,m} = \eta_{\max}\sigma\big((w_g^{\ell,m})^\top \bar{h}_p^\ell + c_g^{\ell,m}\big). \tag{8}$$

Here, U^ℓ and A^ℓ are learnable projections shared across heads, and $\cos(\cdot,\cdot)$ denotes cosine similarity between their ℓ_2-normalized outputs. The parameters $w_g^{\ell,m}$ and $c_g^{\ell,m}$ define the head-specific gate $\eta_p^{\ell,m} \in (0, \eta_{\max})$. For each valid base slot ($\nu_{i,t} = 1$), the backbone attention logit is augmented as

$$s_{pj_i}^{\ell,m} = \frac{(q_p^{\ell,m})^\top k_{j_i}^{\ell,m}}{\sqrt{d_k}} + \eta_p^{\ell,m} b_{pi}^\ell + M_{pj_i}, \tag{9}$$

where $q_p^{\ell,m}$ and $k_{j_i}^{\ell,m}$ are the standard backbone query and key, d_k is the key dimension, and M is the attention mask. The cached identity affects only the additive bias; standard keys and values are computed from the full contextualized tokens. All other logits remain unchanged and are normalized by the same softmax.

3.3 Object-Level Forward Dynamics Module

Action-conditioned object forward dynamics. Let $o_{i,t}$ denote the final VLM hidden state of the object token associated with slot i. An auxiliary head predicts an H_a-step action chunk $\tilde{A}_t$ from mean-pooled, stop-gradient features of valid slots, without using the action expert's output. We construct a position-aware action embedding e_t^a and action conditioning gates $\beta_{i,t}$:

$$e_t^a = \frac{1}{H_a}\sum_{h=1}^{H_a} P_a([\mathrm{sg}(\tilde{A}_{t,h}); p_h]), \qquad \beta_{i,t} = \sigma(P_\beta(\mathrm{sg}(o_{i,t}))). \tag{10}$$

Here, p_h is a learned temporal embedding, P_a and P_β are trainable networks, σ is sigmoid, and sg denotes stop-gradient. The action conditioning gate $\beta_{i,t}$ is distinct from the slot importance gate. During warmup, demonstrated actions $A_t^\star$ replace $\tilde{A}_t$ in the conditioning path. For valid slots, current states are decoded directly, while future prediction is restricted to non-robot object slots:

$$\hat{Y}_{i,t}^{\mathrm{cur}} = D_{\mathrm{cur}}(o_{i,t}), \qquad \hat{Y}_{i,t+1:t+H_f}^{\mathrm{fut}} = D_{\mathrm{fut}}(o_{i,t} + \gamma\beta_{i,t}e_t^a). \tag{11}$$

Here, D_{cur} decodes current states, γ is a fixed residual scale, and D_{fut} predicts all H_f future horizons in parallel, producing horizon-indexed geometry, visibility, and mask logits; robot slots are excluded since their motion is represented by the action chunk. The auxiliary branch is skipped at inference, when the policy directly generates action chunks.

3.4 Training Objectives

Object encoder pretraining. We pretrain E_ω using structured object-level supervision:

$$\mathcal{L}_{\mathrm{enc}} = \lambda_{\mathrm{mask}}\mathcal{L}_{\mathrm{mask}} + \lambda_{\mathrm{geo}}\mathcal{L}_{\mathrm{geo}} + \lambda_{\mathrm{vis}}\mathcal{L}_{\mathrm{vis}} + \lambda_{\mathrm{rt}}\mathcal{L}_{\mathrm{rt}}. \tag{12}$$

Here, $\mathcal{L}_{\mathrm{mask}}$ combines binary cross-entropy and Dice losses for object segmentation. $\mathcal{L}_{\mathrm{geo}}$ supervises object geometry, including boxes, centers, areas, mask–geometry consistency, and auxiliary slot attributes. $\mathcal{L}_{\mathrm{vis}}$ supervises visibility labels derived from offline annotations; visibility is distinct from slot validity. $\mathcal{L}_{\mathrm{rt}}$ is a bidirectional, multi-positive region–text contrastive loss between projected regional visual features and frozen CLIP text features.

VLA training. During VLA training, we keep the pretrained object encoder E_ω frozen and optimize the remaining trainable components with action imitation and auxiliary losses:

$$\mathcal{L} = w_{\mathrm{act}}\mathcal{L}_{\mathrm{act}} + \mathcal{L}_{\mathrm{aux}}, \qquad \mathcal{L}_{\mathrm{aux}} = \mathcal{L}_{\mathrm{cur}} + \mathcal{L}_{\mathrm{fut}} + \mathcal{L}_{\mathrm{prop}} + \mathcal{L}_{\mathrm{gate}}. \tag{13}$$

Here, $\mathcal{L}_{\mathrm{act}}$ is the flow-matching action loss, which also optimizes the prospective object tokens through the action objective. $\mathcal{L}_{\mathrm{cur}}$ supervises current object geometry, visibility, and mask logits, while $\mathcal{L}_{\mathrm{fut}}$ supervises future object states predicted by the forward dynamics module. $\mathcal{L}_{\mathrm{prop}}$ trains the auxiliary coarse action proposal against the demonstrated action chunk, while $\mathcal{L}_{\mathrm{gate}}$ regularizes the learned slot-importance gates used for object-slot prioritization. The routing projections and gates are optimized through the existing objectives without an additional routing loss. $\mathcal{L}_{\mathrm{fut}}$ is computed only for valid non-robot object slots, excluding robot, invalid, and padded slots.

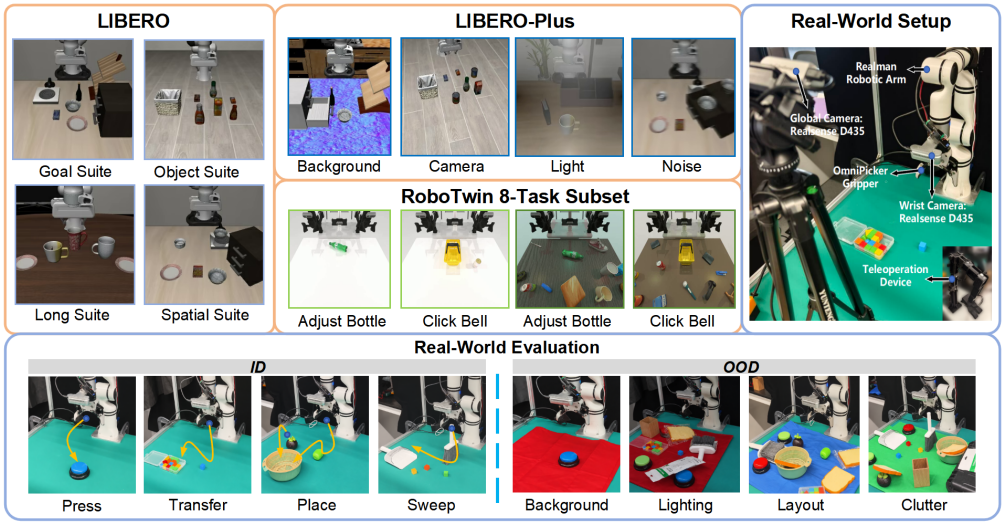


Figure 3: **Experimental overview.** We evaluate ObjDream on LIBERO, LIBERO-Plus, a fixed 8-task subset of RoboTwin 2.0, and four real-world manipulation tasks, covering standard and perturbed evaluation settings.

4 Experiments

Benchmarks. We evaluate ObjDream on LIBERO, LIBERO-Plus, and RoboTwin 2.0, covering both single-arm and bimanual manipulation under diverse distribution shifts. For LIBERO, we follow the standard protocol over the Spatial, Object, Goal, and Long suites (Liu et al., 2023). We further evaluate zero-shot robustness on LIBERO-Plus under seven controlled shifts in camera viewpoint, robot initialization, object layout, language, lighting, background, and sensor noise (Fei et al., 2026). For RoboTwin 2.0, we evaluate on a fixed 8-task subset: *adjust bottle*, *click bell*, *grab roller*, *move can pot*, *open microwave*, *place can basket*, *scan object*, and *turn switch*. These tasks cover diverse manipulation behaviors and object-state transitions. We report results under both Easy and Hard settings (Chen et al., 2025b).

Implementation, training, and evaluation protocol. We use a Qwen3-VL-4B (Bai et al., 2025) backbone and a layerwise flow-matching action expert, following the StarVLA training recipe (Community, 2026). Training proceeds in three stages: (1) we pretrain the spatio-temporal object encoder using offline object-level supervision; (2) with the encoder frozen, we jointly optimize the VLM, action expert, and policy-side object modules using Eq. 13; and (3) we freeze the VLM and object-related modules and refine only the action expert with $\mathcal{L}_{\text{act}}$. Training is conducted on 8 NVIDIA H100 GPUs, while policy inference during evaluation is performed on NVIDIA RTX 4090 GPUs. Each LIBERO task is evaluated over 50 episodes, with the same checkpoint evaluated on LIBERO-Plus without additional fine-tuning. For RoboTwin 2.0, the policy is trained on 50 clean demonstrations per task and evaluated over 100 rollouts per task under both Easy and Hard settings.

4.1 Main Results

LIBERO and LIBERO-Plus. Standard LIBERO measures nominal manipulation performance, while LIBERO-Plus evaluates the same LIBERO-trained policy under controlled distribution shifts without additional fine-tuning. As shown in Table 1, ObjDream achieves the highest average success rate on LIBERO (98.6%) and ranks second overall on LIBERO-Plus (84.4%), with the best results under language (90.6%), lighting (98.4%), and background (98.5%) perturbations and second-best performance under sensor noise (88.6%). These results indicate strong robustness to semantic and appearance variations, while camera viewpoint and robot initialization remain the main challenges.

RoboTwin 2.0. As shown in Fig. 4, we evaluate ObjDream on the fixed eight-task RoboTwin 2.0 subset under Easy and Hard settings. ObjDream achieves average success rates of 60.25% on Easy and 22.75% on Hard, demonstrating competitive robustness under randomized scene variations

Table 1: **Performance on standard LIBERO and zero-shot robustness on LIBERO-Plus.** All LIBERO-Plus results are evaluated without fine-tuning on perturbed environments. Best results are shown in bold, and second-best results are underlined.

Method	LIBERO					LIBERO-Plus							
	Spatial	Object	Goal	Long	Avg	Cam.	Robot	Layout	Lang.	Light	BG	Noise	Overall
General VLA models													
OpenVLA (Kim et al., 2024)	84.7	88.4	79.2	53.7	76.5	0.8	3.5	28.5	23.0	8.1	34.8	15.2	15.6
OpenVLA-OFT (Kim et al., 2025)	97.6	98.4	97.9	94.5	97.1	56.4	31.9	74.2	79.5	88.7	93.3	75.8	69.6
UniVLA (Bu et al., 2025)	96.5	96.8	95.6	92.0	95.2	1.8	46.2	31.9	69.6	69.0	81.0	21.2	42.9
π_0 (Black et al., 2024)	96.8	98.8	95.8	85.2	94.2	13.8	6.0	68.9	58.8	85.0	81.4	79.0	53.6
$\pi_{0.5}$ (Physical Intelligence et al., 2025)	98.8	98.2	98.0	92.4	96.9	75.4	77.5	**85.7**	85.6	96.9	94.6	**89.7**	**85.7**
AVA-VLA (Xiao et al., 2026)	97.4	99.4	97.4	**97.6**	98.0	55.5	25.9	74.1	85.6	95.5	88.9	78.0	70.1
X-VLA (Zheng et al., 2026)	98.2	98.6	97.8	**97.6**	98.1	23.4	**89.7**	71.8	75.7	88.2	96.0	62.7	71.4
Object-centric and predictive models													
WorldVLA (Cen et al., 2025b)	87.6	96.2	83.4	60.0	81.8	0.1	27.9	38.0	41.6	43.7	17.1	10.9	25.0
VLA-JEPA (Sun et al., 2026)	96.2	**99.6**	97.2	95.8	97.2	63.3	67.1	85.1	85.4	95.6	93.6	66.3	79.5
OA-WAM (Liu et al., 2026)	98.9	99.0	97.4	95.9	97.8	**80.5**	89.6	82.8	85.3	96.5	95.9	75.6	83.9
ObjDream (Ours)	**99.2**	**99.6**	**98.2**	97.4	**98.6**	66.2	75.8	81.2	**90.6**	**98.4**	**98.5**	88.6	84.4

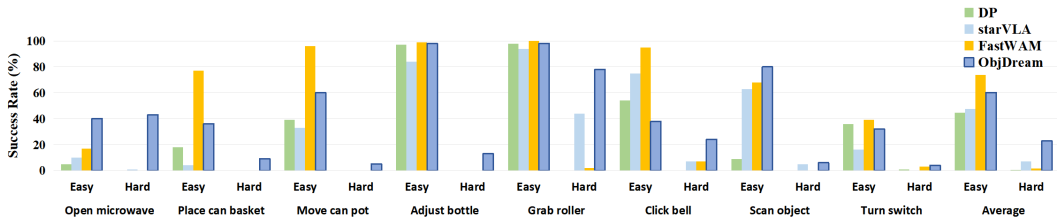


Figure 4: Success rates on the fixed 8-task RoboTwin 2.0 subset under Easy and Hard settings.

without large-scale pretraining on robot data. Compared with StarVLA, ObjDream improves the average success rate by 12.87 percentage points on Easy and 15.62 percentage points on Hard, with a larger gain under the more challenging Hard setting. The improvement is particularly evident on *grab roller*, *open microwave*, and *click bell* under Hard conditions. These results indicate improved robustness to substantial distribution shifts in bimanual manipulation.

4.2 Ablation and Analysis

We conduct ablations on LIBERO-Long and LIBERO-Plus to assess the object interface, object relation modeling (ObjRel), object dynamics (ObjDyn), factorized object injection (Fact. Injection), and identity-guided routing (Id. Routing). Table 2 includes a backbone baseline without the object interface, while "w/o ObjRel & ObjDyn" retains the object tokens, factorized injection, identity-guided routing, and current state supervision.

The full model outperforms the backbone baseline by 2.0 and 3.2 percentage points on LIBERO-Long and LIBERO-Plus, respectively. The object interface without ObjRel or ObjDyn provides a modest improvement over the backbone on LIBERO-Plus and comparable performance on LIBERO-Long. Among single-component removals, disabling factorized injection or identity-guided routing produces the largest decreases in LIBERO-Plus performance, from 84.4% to 82.5% and 82.6%, respectively. Removing ObjDyn reduces the overall success rate to 83.2%, while removing ObjRel results in a smaller decrease to 84.1%. Overall, factorized injection, identity-guided routing, and future object state supervision provide complementary gains.

4.3 Real-World Experiments

Real-world experiments. We evaluate ObjDream on four real-world tasks: *button pressing*, *block pick-and-place*, *two-fruit placement*, and *tabletop sweeping*, covering contact interaction, object manipulation, and long-horizon control. We compare against $\pi_{0.5}$ (Physical Intelligence et al., 2025) and ControlVLA (Li et al., 2025a). All methods use the same task demonstrations for adaptation and are evaluated under in-distribution (ID) and out-of-distribution (OOD) settings without additional fine-tuning.

Table 2: **Ablation study of object-centric components.** Success rates (%) on LIBERO-Long and LIBERO-Plus. Overall is the trial-weighted LIBERO-Plus average. Best results are shown in bold, and second-best results are underlined.

Variant	LIBERO	LIBERO-Plus							
	Long	Cam.	Robot	Layout	Lang.	Light	BG	Noise	Overall
Backbone baseline	95.4	66.4	74.7	80.0	87.1	91.7	92.4	82.8	81.2
w/o ObjRel & ObjDyn	95.2	65.6	73.7	78.9	87.4	95.6	96.9	85.0	82.1
w/o ObjDyn	96.2	66.6	75.0	80.6	88.2	96.2	96.4	87.3	83.2
w/o ObjRel	96.6	**67.4**	**76.1**	80.9	88.2	97.6	97.8	**88.9**	84.1
w/o Fact. Injection	96.2	65.9	75.3	81.0	89.0	94.4	93.5	85.5	82.5
w/o Id. Routing	95.8	65.3	74.9	80.5	87.2	95.3	96.4	86.6	82.6
ObjDream (Full)	**97.4**	66.2	75.8	**81.2**	**90.6**	**98.4**	**98.5**	88.6	**84.4**

Table 3: **Real-robot manipulation results.** Success rates (%) on four real-world tasks under ID and OOD evaluation. Best and second-best results within each setting are bold and underlined.

Eval.	Method	Button Pressing	Block Pick-and-Place	Two-Fruit Placement	Tabletop Sweeping	Avg. Success
ID	$\pi_{0.5}$ (Physical Intelligence et al., 2025)	**90**	**70**	40	65	66
	ControlVLA (Li et al., 2025a)	65	40	**70**	65	60
	ObjDream (Ours)	80	65	60	**90**	**74**
OOD	$\pi_{0.5}$ (Physical Intelligence et al., 2025)	**65**	**60**	40	45	53
	ControlVLA (Li et al., 2025a)	40	10	35	40	31
	ObjDream (Ours)	**65**	50	**50**	**55**	**55**

Results. As shown in Table 3, ObjDream achieves average success rates of 74% and 55% under ID and OOD conditions, respectively, exceeding the strong $\pi_{0.5}$ baseline by 8 percentage points under ID while achieving comparable aggregate performance under OOD. Compared with ControlVLA, ObjDream improves average success rates by 14 and 24 percentage points under ID and OOD conditions, respectively. Notably, under OOD conditions, ObjDream outperforms the best baseline by 10 percentage points on both *two-fruit placement* and *tabletop sweeping*, two tasks that require precise manipulation. These results demonstrate competitive generalization in real-world manipulation without large-scale pretraining on robot data.

5 Conclusion

We present ObjDream, an object-centric VLA framework combining structured object representations with action-conditioned future object state supervision. ObjDream represents objects with persistent identity cues and time-varying state features, and leverages identity-guided routing to direct VLM attention toward task-relevant object tokens. It further supervises object state evolution during training without requiring the decoding of future object states at inference. Experiments on LIBERO, LIBERO-Plus, a fixed 8-task RoboTwin 2.0 subset, and four real-world tasks demonstrate strong manipulation performance and robustness to scene variations.

References

Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zholus, et al. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025.

Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, et al. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025.

Yongjie Bai, Hanting Wang, Mingtong Dai, Qijun Zhong, Yang Liu, and Liang Lin. Bridge-wa: Predicting where and how the world changes for robotic action. *arXiv preprint arXiv:2607.02195*, 2026.

Rokas Bendikas, Daniel Dijkman, Markus Peschl, Sanjay Haresh, and Pietro Mazzaglia. Focusing on What Matters: Object-Agent-Centric Tokenization for Vision Language Action Models. In *Proceedings of The 9th Conference on Robot Learning*, volume 305, pages 3869–3887, 2025.

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0: A Vision-Language-Action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Jasmine Hsu, et al. Rt-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*, 2022.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.

Qingwen Bu, Yanting Yang, Jisong Cai, Shenyuan Gao, Guanghui Ren, Maoqing Yao, Ping Luo, and Hongyang Li. Univla: Learning to act anywhere with task-centric latent actions. *arXiv preprint arXiv:2505.06111*, 2025.

Rui Cai, Jun Guo, Xinze He, Piaopiao Jin, Jie Li, Bingxuan Lin, Futeng Liu, Wei Liu, Fei Ma, Kun Ma, et al. Xiaomi-robotics-0: An open-sourced vision-language-action model with real-time execution. *arXiv preprint arXiv:2602.12684*, 2026.

Jun Cen, Siteng Huang, Yuqian Yuan, Kehan Li, Hangjie Yuan, Chaohui Yu, Bohan Hou, Yuming Jiang, Jiayan Guo, Xin Li, et al. Rynnvla-002: A unified vision-language-action and world model. *arXiv preprint arXiv:2511.17502*, 2025a.

Jun Cen, Chaohui Yu, Hangjie Yuan, Yuming Jiang, Siteng Huang, Jiayan Guo, Xin Li, Yibing Song, Hao Luo, Fan Wang, et al. Worldvla: Towards autoregressive action world model. *arXiv preprint arXiv:2506.21539*, 2025b.

Haonan Chen, Jingxiang Guo, Bangjun Wang, Tianrui Zhang, Xuchuan Huang, Boren Zheng, Yiwen Hou, Chenrui Tie, Jiajun Deng, and Lin Shao. Goal-vla: Image-generative vlms as object-centric world models empowering zero-shot robot manipulation. *arXiv preprint arXiv:2506.23919*, 2025a.

Tianxing Chen, Zanxin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Zixuan Li, Qiwei Liang, Xianliang Lin, Yiheng Ge, Zhenyu Gu, et al. Robotwin 2.0: A scalable data generator and benchmark with strong domain randomization for robust bimanual robotic manipulation. *arXiv preprint arXiv:2506.18088*, 2025b.

Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 44(10-11):1684–1704, 2025.

Nhat Chung, Taisei Hanyu, Toan Nguyen, Huy Le, Frederick Bumgarner, Duy Minh Ho Nguyen, Khoa Vo, Kashu Yamazaki, Chase Rainwater, Tung Kieu, Anh Nguyen, and Ngan Le. Rethinking progression of memory state in robotic manipulation: An object-centric perspective. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(5):3407–3415, 2026. doi: 10.1609/aaai.v40i5.37337.

StarVLA Community. Starvla: A lego-like codebase for vision-language-action model developing. *arXiv preprint arXiv:2604.05014*, 2026.

Yilun Du, Sherry Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Josh Tenenbaum, Dale Schuurmans, and Pieter Abbeel. Learning universal policies via text-guided video generation. *Advances in neural*

information processing systems, 36:9156–9172, 2023.

Senyu Fei, Siyin Wang, Junhao Shi, Zihao Dai, Jikun Cai, Pengfang Qian, Li Ji, Xinzhe He, Shiduo Zhang, Zhaoye Fei, Jinlan Fu, Jingjing Gong, and Xipeng Qiu. Libero-plus: A progressive robustness benchmark for visual-language-action models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 38574–38583, June 2026.

Stefano Ferraro, Pietro Mazzaglia, Tim Verbelen, and Bart Dhoedt. Focus: object-centric world models for robotic manipulation. *Frontiers in Neurorobotics*, 19:1585386, 2025.

Xinyu Guo, Bin Xie, Wei Chai, Xianchi Deng, Tiancai Wang, Zhengxing Wu, and Xingyu Chen. Priorvla: Prior-preserving adaptation for vision-language-action models. *arXiv preprint arXiv:2605.10925*, 2026.

Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International conference on machine learning*, pages 2555–2565. PMLR, 2019.

Asher Hancock, Xindi Wu, Lihan Zha, Olga Russakovsky, and Anirudha Majumdar. Actions as language: Fine-tuning vlms into vlas without catastrophic forgetting. In *International Conference on Learning Representations*, volume 2026, pages 75333–75360, 2026.

Taisei Hanyu, Nhat Chung, Huy Le, Toan Nguyen, Yuki Ikebe, Anthony Gunderman, Duy Nguyen Ho Minh, Khoa Vo, Tung Kieu, Kashu Yamazaki, et al. Slotvla: Towards modeling of object-relation representations in robotic manipulation. *arXiv preprint arXiv:2511.06754*, 2025.

Zhi Hou, Tianyi Zhang, Yuwen Xiong, Haonan Duan, Hengjun Pu, Ronglei Tong, Chengyang Zhao, Xizhou Zhu, Yu Qiao, Jifeng Dai, et al. Dita: Scaling diffusion transformer for generalist vision-language-action policy. In *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7686–7697. IEEE, 2025.

Youngjoon Jeong, Junha Chun, Soonwoo Cha, and Taesup Kim. Object-centric world model for language-guided manipulation. *arXiv preprint arXiv:2503.06170*, 2025.

Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.

Moo Jin Kim, Chelsea Finn, and Percy Liang. Fine-tuning vision-language-action models: Optimizing speed and success. *arXiv preprint arXiv:2502.19645*, 2025.

Thomas Kipf, Gamaleldin F Elsayed, Aravindh Mahendran, Austin Stone, Sara Sabour, Georg Heigold, Rico Jonschkowski, Alexey Dosovitskiy, and Klaus Greff. Conditional object-centric learning from video. *arXiv preprint arXiv:2111.12594*, 2021.

Fuhao Li, Wenxuan Song, Han Zhao, Jingbo Wang, Pengxiang Ding, Donglin Wang, Long Zeng, and Haoang Li. Spatial forcing: Implicit spatial representation alignment for vision-language-action model. In *International Conference on Learning Representations*, volume 2026, pages 132324–132345, 2026a.

Puhao Li, Yingying Wu, Ziheng Xi, Wanlin Li, Yuzhe Huang, Zhiyuan Zhang, Yinghan Chen, Jianan Wang, Song-Chun Zhu, Tengyu Liu, and Siyuan Huang. ControlVLA: Few-Shot Object-Centric Adaptation for Pre-Trained Vision-Language-Action Models. In *Proceedings of The 9th Conference on Robot Learning*, volume 305, pages 1898–1913, 2025a.

Wei Li, Peijin Jia, Yuan Ma, Xuefeng Jiang, Titong Jiang, Sheng Sun, Yujian Li, Xin Wen, Han Hong, Zhikang Liu, et al. Fomovla: Bridging visual foresight and motion guidance for vision-language-action models. *arXiv preprint arXiv:2607.14739*, 2026b.

Xiang Li, Yupeng Zheng, Songen Gu, Huailiang Ma, Feng Yu, Xian Nie, Shanshuai Yuan, Yujie Zang, Weize Li, Shuai Tian, et al. Latent action as intention enables efficient future imagination for world action models. *arXiv preprint arXiv:2608.24882*, 2026c.

Xiaoqi Li, Jingyun Xu, Mingxu Zhang, Jiaming Liu, Yan Shen, Iaroslav Ponomarenko, Jiahui Xu, Liang Heng, Siyuan Huang, Shanghang Zhang, et al. Object-centric prompt-driven vision-language-action model for robotic manipulation. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 27638–27648. IEEE, 2025b.

Yi Li, Yuquan Deng, Jesse Zhang, Joel Jang, Marius Memmel, Caelan Garrett, Fabio Ramos, Dieter Fox, Anqi Li, Abhishek Gupta, et al. Hamster: Hierarchical action models for open-world robot

manipulation. In *International Conference on Learning Representations*, volume 2025, pages 24040–24068, 2025c.

Wenqi Liang, Gan Sun, Yao He, Jiahua Dong, Suyan Dai, Ivan Laptev, Salman Khan, and Yang Cong. Pixelvla: Advancing pixel-level understanding in vision-language-action model. In *International Conference on Learning Representations*, volume 2026, pages 110464–110481, 2026.

Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. *arXiv preprint arXiv:2210.02747*, 2022.

Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. Libero: Benchmarking knowledge transfer for lifelong robot learning. *Advances in Neural Information Processing Systems*, 36:44776–44791, 2023.

Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, et al. Grounding dino: Marrying dino with grounded pre-training for open-set object detection. In *European conference on computer vision*, pages 38–55. Springer, 2024.

Yushan Liu, Peibo Sun, Shoujie Li, Yifan Xie, Lingfeng Zhang, Xintao Chao, Shiyuan Dong, Fang Chen, Xiao-Ping Zhang, and Wenbo Ding. Oa-wam: Object-addressable world action model for robust robot manipulation. *arXiv preprint arXiv:2605.06481*, 2026.

Francesco Locatello, Dirk Weissenborn, Thomas Unterthiner, Aravindh Mahendran, Georg Heigold, Jakob Uszkoreit, Alexey Dosovitskiy, and Thomas Kipf. Object-centric learning with slot attention. *Advances in neural information processing systems*, 33:11525–11538, 2020.

Malte Mosbach, Jan Niklas Ewertz, Angel Villar-Corrales, and Sven Behnke. Sold: Slot object-centric latent dynamics models for relational manipulation learning from pixels. *arXiv preprint arXiv:2410.08822*, 2024.

NVIDIA, Johan Bjorck, Fernando Castaneda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, et al. GR00T N1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734*, 2025.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

Abby O'Neill, Abdul Rehman, Abhiram Maddukuri, Abhishek Gupta, Abhishek Padalkar, Abraham Lee, Acorn Pooley, Agrim Gupta, Ajay Mandlekar, Ajinkya Jain, et al. Open x-embodiment: Robotic learning datasets and rt-x models. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 6892–6903. IEEE, 2024.

Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, et al. $\pi_{0.5}$: A vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025.

Delin Qu, Haoming Song, Qizhi Chen, Yuanqi Yao, Xinyi Ye, Yan Ding, Zhigang Wang, JiaYuan Gu, Bin Zhao, Dong Wang, et al. Spatialvla: Exploring spatial representations for visual-language-action model. *arXiv preprint arXiv:2501.15830*, 2025.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 8748–8763. PMLR, 2021.

Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. In *International Conference on Learning Representations*, volume 2025, pages 28085–28128, 2025.

Jingwen Sun, Wenyao Zhang, Zekun Qi, Shaojie Ren, Zezhi Liu, Hanxin Zhu, Guangzhong Sun, Xin Jin, and Zhibo Chen. Vla-jepa: Enhancing vision-language-action model with latent world model. *arXiv preprint arXiv:2602.10098*, 2026.

Octo Model Team, Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, et al. Octo: An open-source generalist robot policy. *arXiv preprint arXiv:2405.12213*, 2024.

Xiaomi Robotics Team, Jun Guo, Piaopiao Jin, Jason Li, Peiyan Li, Yingyan Li, Futeng Liu, Wanli Peng, Optimus Qin, Yifei Su, et al. Xiaomi-robotics-1: Scaling vision-language-action models with over 100k hours of real-world trajectories. *arXiv preprint arXiv:2607.15330*, 2026.

Sieu Tran, Duc Nguyen, Hao Vo, Khoa Vo, and Ngan Le. Dual-state slot attention: Decoupling appearance and identity for video object-centric learning. *arXiv preprint arXiv:2606.12601*, 2026.

Angel Villar-Corrales and Sven Behnke. Playslot: Learning inverse latent dynamics for controllable object-centric video prediction and planning. *arXiv preprint arXiv:2502.07600*, 2025.

Khoa Vo, Taisei Hanyu, Yuki Ikebe, Trong Thang Pham, Nhat Chung, Minh Nhat Vu, Duy Nguyen Ho Minh, Anh Nguyen, Anthony Gunderman, Chase Rainwater, et al. Clutter-robust vision-language-action models through object-centric and geometry grounding. *arXiv preprint arXiv:2512.22519*, 2025.

Kuanning Wang, Ke Fan, Chenhao Qiu, Zeyu Shangguan, Yuqian Fu, Yanwei Fu, Daniel Seita, and Xiangyang Xue. Oflow: Injecting object-aware temporal flow matching for robust robotic manipulation. *arXiv preprint arXiv:2604.17876*, 2026a.

Qiuyue Wang, Mingsheng Li, Jian Guan, Jinhui Ye, Sicheng Xie, Yitao Liu, Junhao Chen, Zhixuan Liang, Jie Zhang, Xintong Hu, et al. Qwen-vla: Unifying vision-language-action modeling across tasks, environments, and robot embodiments. *arXiv preprint arXiv:2605.30280*, 2026b.

Xinkai Wang, Chenyi Wang, Yifu Xu, Mingzhe Ye, Fu-Cheng Zhang, Jialin Tian, Xinyu Zhan, Lifeng Zhu, Cewu Lu, and Lixin Yang. Lamp: Learning vision-language-action policies with 3d scene flow as latent motion prior. *arXiv preprint arXiv:2603.25399*, 2026c.

Jiude Wei, Yuxuan Li, Cewu Lu, and Jianhua Sun. Physically ground commonsense knowledge for articulated object manipulation with analytic concepts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13519–13528, 2026.

Philipp Wu, Alejandro Escontrela, Danijar Hafner, Pieter Abbeel, and Ken Goldberg. Daydreamer: World models for physical robot learning. In *Conference on robot learning*, pages 2226–2240. PMLR, 2023.

Wei Wu, Fan Lu, Yunnan Wang, Shuai Yang, Shi Liu, Fangjing Wang, Qian Zhu, He Sun, Yong Wang, Shuailei Ma, et al. A pragmatic vla foundation model. *arXiv preprint arXiv:2601.18692*, 2026a.

Wei Wu, Fangjing Wang, Fan Lu, He Sun, Shi Liu, Yunnan Wang, Yibin Yan, Yong Wang, Shuailei Ma, Xinyang Wang, et al. From foundation to application: Improving vla models in practice. *arXiv preprint arXiv:2607.06403*, 2026b.

Ziyi Wu, Nikita Dvornik, Klaus Greff, Thomas Kipf, and Animesh Garg. Slotformer: Unsupervised visual dynamics simulation with object-centric models. *arXiv preprint arXiv:2210.05861*, 2022.

Lei Xiao, Jifeng Li, Juntao Gao, Feiyang Ye, Yan Jin, Jingjing Qian, Jing Zhang, Yong Wu, and Xiaoyuan Yu. Ava-vla: Improving vision-language-action models with active visual attention. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13453–13463, 2026.

Yue Yang, Shuo Cheng, Yu Fang, Homanga Bharadhwaj, Mingyu Ding, Gedas Bertasius, and Daniel Szafir. Lilo-vla: Compositional long-horizon manipulation via linked object-centric policies. *arXiv preprint arXiv:2602.21531*, 2026.

Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, et al. World action models are zero-shot policies. *arXiv preprint arXiv:2602.15922*, 2026.

Hanyang Yu, Haitao Lin, Jingbo Zhang, Wenyao Zhang, Chenghao Gu, Heng Li, and Ping Tan. Maskwam: Unifying mask prompting and prediction for world-action models. *arXiv preprint arXiv:2606.13515*, 2026a.

Qize Yu, Jiadi You, Yuran Wang, Jiaqi Liang, Bowen Ping, Yang Tian, Yue Chen, Minghong Cai, Zeying Gong, Ruihai Wu, et al. Affordancevla: A vision-language-action model empowering action generation through affordance-aware understanding. *arXiv preprint arXiv:2606.06155*, 2026b.

Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-wam: Do world action models need test-time future imagination? *arXiv preprint arXiv:2603.16666*, 2026.

Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, XinQiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. DreamVLA: A Vision-Language-Action Model Dreamed with Comprehensive World Knowledge. In *Advances in Neural Information Processing Systems*, volume 38, pages 24195–24228, 2025.

Qingqing Zhao, Yao Lu, Moo Jin Kim, Zipeng Fu, Zhuoyang Zhang, Yecheng Wu, Zhaoshuo Li, Qianli Ma, Song Han, Chelsea Finn, et al. Cot-vla: Visual chain-of-thought reasoning for vision-language-action models. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1702–1713. IEEE, 2025.

Tony Z Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. *arXiv preprint arXiv:2304.13705*, 2023.

Jinliang Zheng, Jianxiong Li, Zhihao Wang, Dongxiu Liu, Xirui Kang, Yuchun Feng, Yinan Zheng, Jiayin Zou, Yilun Chen, Jia Zeng, et al. X-vla: Soft-prompted transformer as scalable cross-embodiment vision-language-action model. In *International Conference on Learning Representations*, volume 2026, pages 60580–60606, 2026.

Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. Dino-wm: World models on pre-trained visual features enable zero-shot planning. *arXiv preprint arXiv:2411.04983*, 2024a.

Siyuan Zhou, Yilun Du, Jiaben Chen, Yandong Li, Dit-Yan Yeung, and Chuang Gan. Robodreamer: Learning compositional world models for robot imagination. *arXiv preprint arXiv:2404.12377*, 2024b.

Yifeng Zhu, Abhishek Joshi, Peter Stone, and Yuke Zhu. Viola: Imitation learning for vision-based manipulation with object proposal priors. In *Conference on Robot Learning*, pages 1199–1210. PMLR, 2023.

A Offline Annotation Pipeline

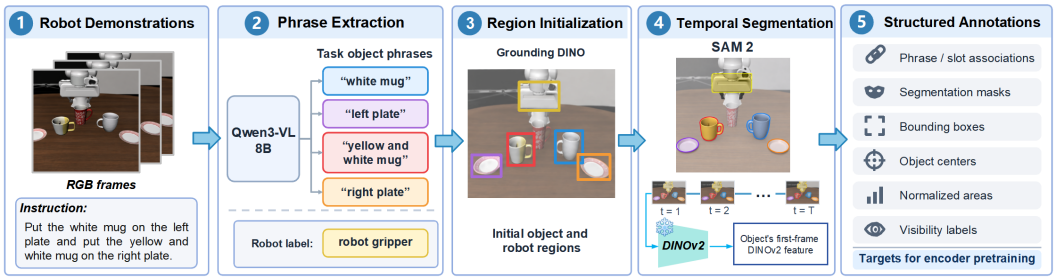


Figure 5: **Automatic offline annotation pipeline for object encoder pretraining.** Qwen3-VL-8B extracts task-relevant object phrases from demonstration instructions and initial observations, while a separate *robot gripper* label is used for robot annotation. Grounding DINO initializes the corresponding object regions, SAM2 propagates segmentation masks across frames, and DINOv2 extracts object-specific features from the initial grounded regions. Masks, geometry, visibility, and phrase–region correspondences are automatically constructed from this pipeline without manual frame-wise object annotation. The resulting annotations supervise encoder pretraining, while the DINOv2 features provide visual identity references.

Importantly, the object-level supervision is automatically generated from the demonstration trajectories and does not require manual frame-wise annotation of masks, boxes, geometry, or visibility. For each trajectory with instruction x and visual observations, we construct object-level supervision through phrase extraction, grounded localization, mask propagation, and automatic object-state annotation, yielding temporally aligned masks, geometry, visibility, and phrase–region correspondences. The offline annotation pipeline is illustrated in Fig. 5.

Instruction-conditioned object phrases. During automatic offline annotation, Qwen3-VL-8B (Bai et al., 2025) takes the language instruction x together with the initial visual observation I_0 and extracts the task-relevant phrase set P_0. The extracted phrases are fixed for the entire trajectory and used as consistent semantic references across frames. When multiple instances share the same category, coarse spatial cues, such as relative left/right positions, are included when needed to distinguish them. During offline annotation, the label *robot gripper* is used to localize robot regions and assign their robot role.

Phrase-grounded localization and temporal segmentation. Given P_0, Grounding DINO (Liu et al., 2024) provides initial phrase-conditioned localization, and SAM2 (Ravi et al., 2025) propagates the corresponding masks across the demonstration trajectory. The phrase associated with each object remains fixed while its spatial support changes with motion, occlusion, or visibility. Boxes, centers, areas, and visibility labels are automatically derived from the propagated masks and normalized with respect to the image dimensions where applicable. DINOv2 (Oquab et al., 2023) features are pooled over the initial grounded object region to construct the visual identity reference during encoder pretraining. Thus, the masks, geometry, and visibility labels serve as automatically generated offline supervision and are not provided as policy inputs during rollout. Grounding DINO localization and SAM2 video mask propagation are used only for offline annotation and are not executed during rollout.

B Future Object State Prediction Analysis

We analyze whether the auxiliary future object state decoder learns nontrivial object state evolution rather than simply preserving the current state. We evaluate three representative LIBERO manipulation tasks covering object placement and articulated-object interaction. Only valid non-robot object slots are evaluated; robot and padding slots are excluded. For each prediction origin, the decoder predicts object centers, bounding boxes, and masks at every consecutive future step $h \in \{1, \ldots, H_f\}$, with $H_f = 8$.

We compare the learned decoder against a *current state persistence* baseline that repeats the current object estimate at each future step. We report an *Unclamped* subset containing only predictions whose future targets are directly available within the trajectory, and a *Displacement* subset containing object–target pairs whose centroid changes by more than 1% of the image diagonal. Center error is normalized by the image diagonal, while box and mask IoU measure predicted object extent.

Table 4: **Future object state prediction accuracy on LIBERO.** Current state persistence repeats the current object state at each future target. Bold indicates the better value within each comparison.

Subset	**Method**	**Center error (%) ↓**	**Box IoU ↑**	**Mask IoU ↑**
Unclamped	Current state persistence	6.37	0.634	0.564
	Future prediction	**1.60**	**0.690**	**0.616**
Displacement	Current state persistence	13.91	0.287	0.245
	Future prediction	**2.03**	**0.575**	**0.542**

The learned decoder consistently improves over current state persistence. On the Unclamped subset, the learned decoder has lower center error and higher box and mask IoU than current state persistence. The gains are more pronounced under object displacement, where center error decreases from 13.91% to 2.03% and box and mask IoU improve from 0.287 and 0.245 to 0.575 and 0.542. These results indicate that the auxiliary decoder captures nontrivial future object state changes rather than merely propagating the current state.

Qualitative future object state predictions. Figure 6 shows representative LIBERO future object state predictions. Although the decoder predicts all horizons $h = 1, \ldots, H_f$, we display only $h = 1, 4, 8$ for clarity. Figures 8 and 7 show additional predictions on real-world tasks and RoboTwin 2.0, respectively.

Table 5: **Sensitivity of future object state prediction to action conditioning on LIBERO.** *Native Predicted Action* uses the original auxiliary action proposal; *No Action* removes the position-aware action embedding; and *Mismatched Action* replaces it with the position-aware embedding of a predicted action chunk from another sample within the same task. All other model components and prediction targets are kept fixed.

Subset	Conditioning	Center error (%) ↓	Box IoU ↑	Mask IoU ↑
Unclamped	No Action	1.90	0.659	0.609
	Mismatched Action	2.02	0.660	0.587
	Native Predicted Action	**1.60**	**0.690**	**0.616**
Displacement	No Action	2.35	0.546	0.534
	Mismatched Action	2.71	0.532	0.486
	Native Predicted Action	**2.03**	**0.575**	**0.542**

Effect of action conditioning on future object state prediction. To examine whether the decoder uses action information, we intervene on the action conditioning path using the same trained model and evaluation tasks. Model parameters, contextualized object tokens, and action conditioning gates are kept fixed. *Native Predicted Action* uses the original auxiliary action proposal; *No Action* sets the position-aware action embedding to zero; and *Mismatched Action* replaces it with the position-aware embedding of a predicted action chunk from another sample within the same task. As shown in Table 5, the native condition yields the lowest center error and the highest box and mask IoU on both subsets. On the Unclamped subset, center error is 1.60% with the native condition,

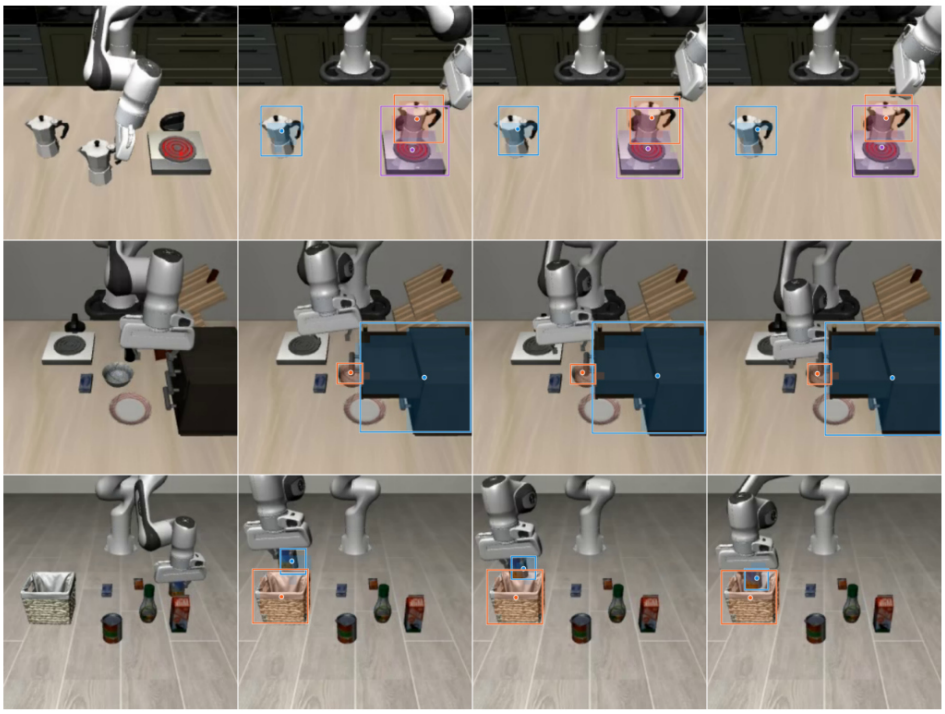


Figure 6: **Qualitative future object mask predictions on LIBERO.** Rows show three representative manipulation tasks. The decoder predicts all future steps $h = 1, \ldots, H_f$ for non-robot objects; for clarity, we visualize only $h = 1, 4, 8$. The predicted masks are overlaid on the corresponding future RGB frames, which are used only for visualization and are not provided to the decoder.



Figure 7: **Qualitative future object state predictions on RoboTwin 2.0.** Rows show representative tasks from the fixed eight-task subset. The decoder predicts all future steps for non-robot objects; for clarity, we visualize masks and bounding boxes only at $h = 1, 4, 8$. The predictions are overlaid on the corresponding future RGB frames, which are used only for visualization and are not provided to the decoder.

compared with 1.90% without action conditioning and 2.02% with mismatched actions. On the Displacement subset, center error increases from 2.03% to 2.35% and 2.71%, respectively. Box IoU decreases from 0.575 to 0.546 and 0.532, while mask IoU decreases from 0.542 to 0.534 and 0.486.

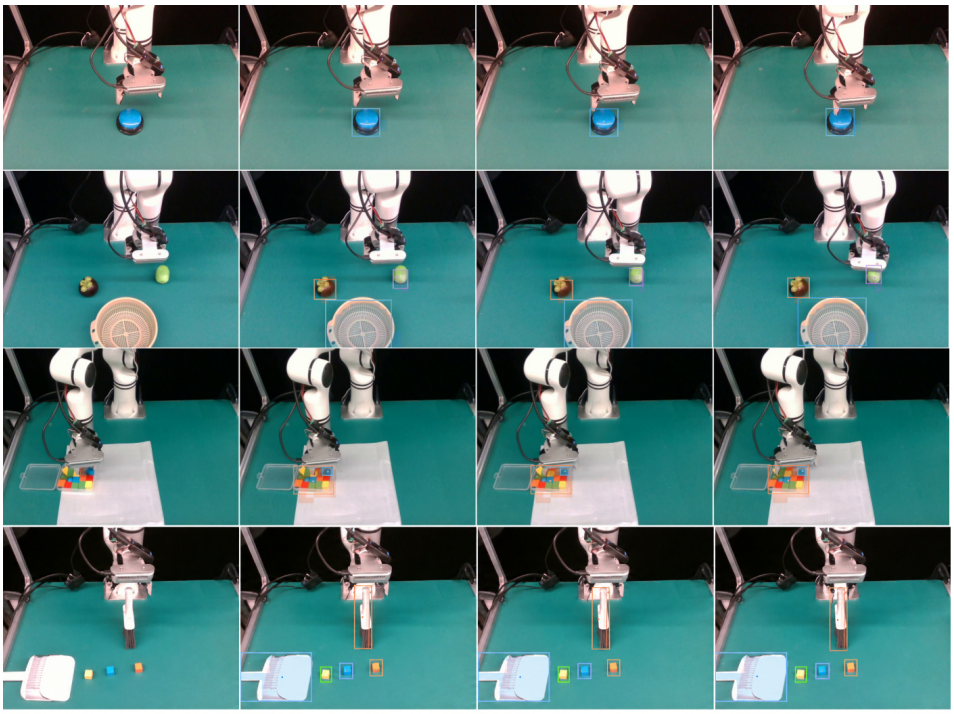


Figure 8: **Qualitative future object state predictions on real-world tasks.** Rows show button pressing, two-fruit placement, block pick-and-place, and tabletop sweeping. The decoder predicts all future steps for non-robot objects; for clarity, we visualize masks and bounding boxes only at $h = 1, 4, 8$. The predictions are overlaid on the corresponding future RGB frames, which are used only for visualization and are not provided to the decoder.

These interventions indicate that the future object state decoder uses the action condition, while not establishing counterfactual dynamics under arbitrary action sequences.